Problem Set 6

Foundations of Machine Learning

- **Submission**: Turn in as a `PDF` and a **separate source code file** on `http://mycourses.unh.edu`. Scanned handwritten notes are encouraged.
- **Questions**: See the class website for a link to office hours

Relevant reading: MML 5.2–5.3 (chain rule, Jacobian); CO 2.1–2.2, 3.1–3.2 (convex sets and functions). Problems marked **[code]** should be submitted as a program in a language of your choice (Julia or Python recommended). The cited exercise numbers are from MML (Deisenroth et al., 2021) and CO (Boyd & Vandenberghe, 2004).

1. MML, Chapter 5 exercises: any two on the chain rule and the Jacobian of a vector-valued function.
2. CO, Chapter 2 exercises: any two on convex sets (e.g. Exercises 2.1 and 2.5).
3. CO, Chapter 3 exercises: any two on convex functions (e.g. Exercises 3.1 and 3.6).
4. **[code]** For a composition $h(x) = g(f(x))$ of two vector-valued functions of your choice, compute the Jacobian by the chain rule, then verify it numerically by finite differences at a point.
5. **Time spent.** Report approximately how many hours you spent completing this problem set (including the reading).